Contents

Repository

GitHub: https://github.com/Newterios/educationalLMS **Live:** https://aitbek.tech

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Team:

Role	Member
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Course: Site Reliability Engineering **Date:** 2026-05-19

Final Project — Comprehensive SRE Implementation

Distributed microservices system **deployed on `aitbek.tech`** with a full CI/CD pipeline (GitHub Actions), Docker Compose, Docker Swarm, Kubernetes, Terraform (DigitalOcean), Ansible, Prometheus and Grafana.

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1. Repository Structure

repo tree

sre folder

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2. Kubernetes

k8s folder

```
kubectl apply -f sre/k8s/
kubectl get pods,svc,ing -n edulms
```

```
namespace/edulms configured
configmap/edulms-config configured
secret/edulms-secrets configured
persistentvolumeclaim/postgres-pvc configured
deployment.apps/postgres configured
service/postgres unchanged
deployment.apps/redis configured
service/redis unchanged
deployment.apps/auth-service configured
service/auth-service unchanged
deployment.apps/user-service configured
service/user-service unchanged
deployment.apps/course-service configured
service/course-service unchanged
deployment.apps/assessment-service configured
service/assessment-service unchanged
deployment.apps/attendance-service configured
service/attendance-service unchanged
deployment.apps/payment-service configured
service/payment-service unchanged
deployment.apps/notification-service configured
service/notification-service unchanged
deployment.apps/gateway configured
service/gateway unchanged
deployment.apps/web configured
```

kubectl get pods,svc,ing -n edulms

```
NAME                                        READY   STATUS    RESTARTS   AGE
pod/postgres-66c78786d-j7pzz                1/1     Running   0          3m
pod/redis-849468cd6d-17cq7                  1/1     Running   0          3m
pod/gateway-848bb65f6-djpsm                 1/1     Running   0          3m
pod/gateway-848bb65f6-x8s5t                 1/1     Running   0          3m
pod/auth-service-7567fc565d-lmwlv           1/1     Running   0          2m
pod/auth-service-7567fc565d-n7zwc           1/1     Running   0          2m
pod/user-service-85fdb85c94-4rxkn           1/1     Running   0          2m
pod/user-service-85fdb85c94-hv42r           1/1     Running   0          2m
pod/course-service-77df5fcfb7-2xbrt         1/1     Running   0          2m
pod/course-service-77df5fcfb7-mfbxp         1/1     Running   0          2m
pod/assessment-service-5bdb6d6fb9-87rgg     1/1     Running   0          2m
pod/assessment-service-5bdb6d6fb9-q5r8m     1/1     Running   0          2m
pod/attendance-service-858d88f58c-76z9f     1/1     Running   0          2m
pod/attendance-service-858d88f58c-956vc     1/1     Running   0          2m
pod/payment-service-9c7ccdb44-2jc7g         1/1     Running   0          2m
pod/payment-service-9c7ccdb44-9nwr2         1/1     Running   0          2m
pod/notification-service-86b48fbb74-26c8c   1/1     Running   0          2m
pod/notification-service-86b48fbb74-29njr   1/1     Running   0          2m
pod/web-c65b996cc-q27cr                     1/1     Running   0          2m

NAME                           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/assessment-service     ClusterIP   10.96.170.189   <none>        8004/TCP   3m
service/attendance-service     ClusterIP   10.96.3.229     <none>        8005/TCP   3m
service/auth-service           ClusterIP   10.96.5.122     <none>        8001/TCP   3m
```

Terraform

Figure 1: kubectl apply

3. Terraform (DigitalOcean)

terraform folder

```
terraform init && terraform plan && terraform apply
```

---

4. Ansible

ansible folder

```
ansible-playbook -i inventory.yml playbooks/01_prepare.yml
ansible-playbook -i inventory.yml playbooks/02_deploy.yml
ansible-playbook -i inventory.yml playbooks/03_monitoring.yml
```

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5. CI/CD Pipeline (new for the team final)

Flow: **branch → push/PR → CI (lint/test/build) → CD (SSH deploy)**

5 stages of CD job:

1. SSH to `aitbek.tech`
2. `git pull` on the server
3. Run Ansible playbook
4. Pull / build Docker images
5. `kubectl apply` + rolling restart + **health check**

cicd new yml

github actions runs

github actions detail

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6. Deployed System — aitbek.tech

website live

server deploy

server docker ps

---

7. Monitoring & Alerting

prometheus targets live

grafana live

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8. Incident Simulation — PostgresDown Alert FIRING

We stopped the `postgres-exporter` container to demonstrate the full alerting pipeline. PostgreSQL itself kept serving traffic.

```
docker stop diplom-postgres-exporter-1
```

main.tf

educationalLMS > terraform > main.tf

```
resource "digitalocean_vpc" "edulms" {
  name     = "${var.project_name}-vpc"
  region   = var.region
  ip_range = var.vpc_ip_range
}

resource "digitalocean_firewall" "edulms" {
  name = "${var.project_name}-fw"
  droplet_ids = [digitalocean_droplet.edulms_app.id]

  inbound_rule {
    protocol         = "tcp"
    port_range       = "22"
    source_addresses = ["0.0.0.0/0", "::/0"]
  }

  inbound_rule {
    protocol         = "tcp"
    port_range       = "80"
    source_addresses = ["0.0.0.0/0", "::/0"]
  }

  inbound_rule {
    protocol         = "tcp"
    port_range       = "443"
    source_addresses = ["0.0.0.0/0", "::/0"]
  }

  inbound_rule {
    protocol         = "icmp"
    source_addresses = ["0.0.0.0/0", "::/0"]
  }
```

Ansible

ansible-playbook -i inventory.yml playbooks/01_prepare.yml

Figure 2: terraform main.tf

```
>> terraform plannal_edulms\educationalLMS>
Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
PS D:\AITU\sre\final_edulms\educationalLMS\terraform> terraform plan
Terraform used the selected providers to generate the following execution plan. Resource actions are
indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # digitalocean_droplet.edulms_app will be created
  + resource "digitalocean_droplet" "edulms_app" {
      + backups              = false
      + created_at           = (known after apply)
      + disk                 = (known after apply)
      + graceful_shutdown    = false
      + id                   = (known after apply)
      + image                = "ubuntu-22-04-x64"
      + ipv4_address         = (known after apply)
      + ipv4_address_private = (known after apply)
      + ipv6                 = false
      + ipv6_address         = (known after apply)
      + locked               = (known after apply)
      + memory               = (known after apply)
      + monitoring           = true
      + name                 = "edulms-app-1"
      + price_hourly         = (known after apply)
      + price_monthly        = (known after apply)
```

Ln 1, Col 1

Figure 3: terraform plan

```
PLAY [Prepare host for EduLMS deployment] ***********************************

TASK [Update apt cache] *****************************************************
changed: [edulms-server]

TASK [Install base dependencies] ********************************************
changed: [edulms-server]

TASK [Ensure Docker service enabled] ****************************************
ok: [edulms-server]

TASK [Ensure project root exists] *******************************************
ok: [edulms-server]

PLAY RECAP ******************************************************************
edulms-server : ok=4  changed=2  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

ansible-playbook -i inventory.yml playbooks/02_deploy.yml

```
PLAY [Deploy EduLMS with Docker Compose] ************************************

TASK [Clone or update repository] *******************************************
changed: [edulms-server]

TASK [Copy environment file if missing] *************************************
ok: [edulms-server]

TASK [Start stack] **********************************************************
changed: [edulms-server]

TASK [Verify containers] ****************************************************
changed: [edulms-server]

TASK [Show compose status] **************************************************
ok: [edulms-server] => {
  "compose_ps.stdout_lines": [
    "NAME                    STATUS",
```

ansible-playbook -i inventory.yml playbooks/03_monitoring.yml

```
PLAY [Configure monitoring stack] *******************************************

TASK [Ensure observability files exist] *************************************
ok: [edulms-server]

TASK [Fail if Prometheus config missing] ************************************
skipping: [edulms-server]

TASK [Restart services to apply monitoring changes] *************************
changed: [edulms-server]

PLAY RECAP ******************************************************************
edulms-server : ok=2  changed=1  unreachable=0  failed=0  skipped=1  rescued=0  ignored=0
```

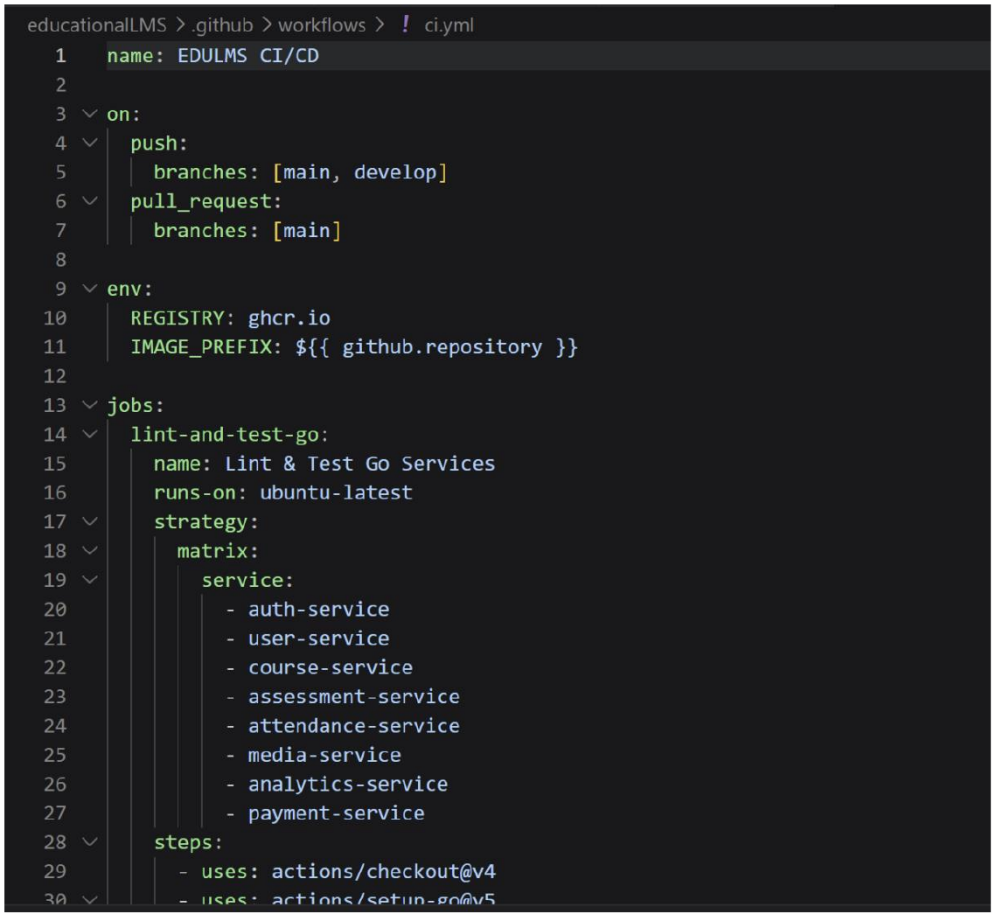
Incident

To demonstrate the full alerting pipeline end-to-end, I intentionally stopped the PostgreSQL exporter

container to simulate a monitoring component failure:

How I triggered the PostgresDown FIRING alert:

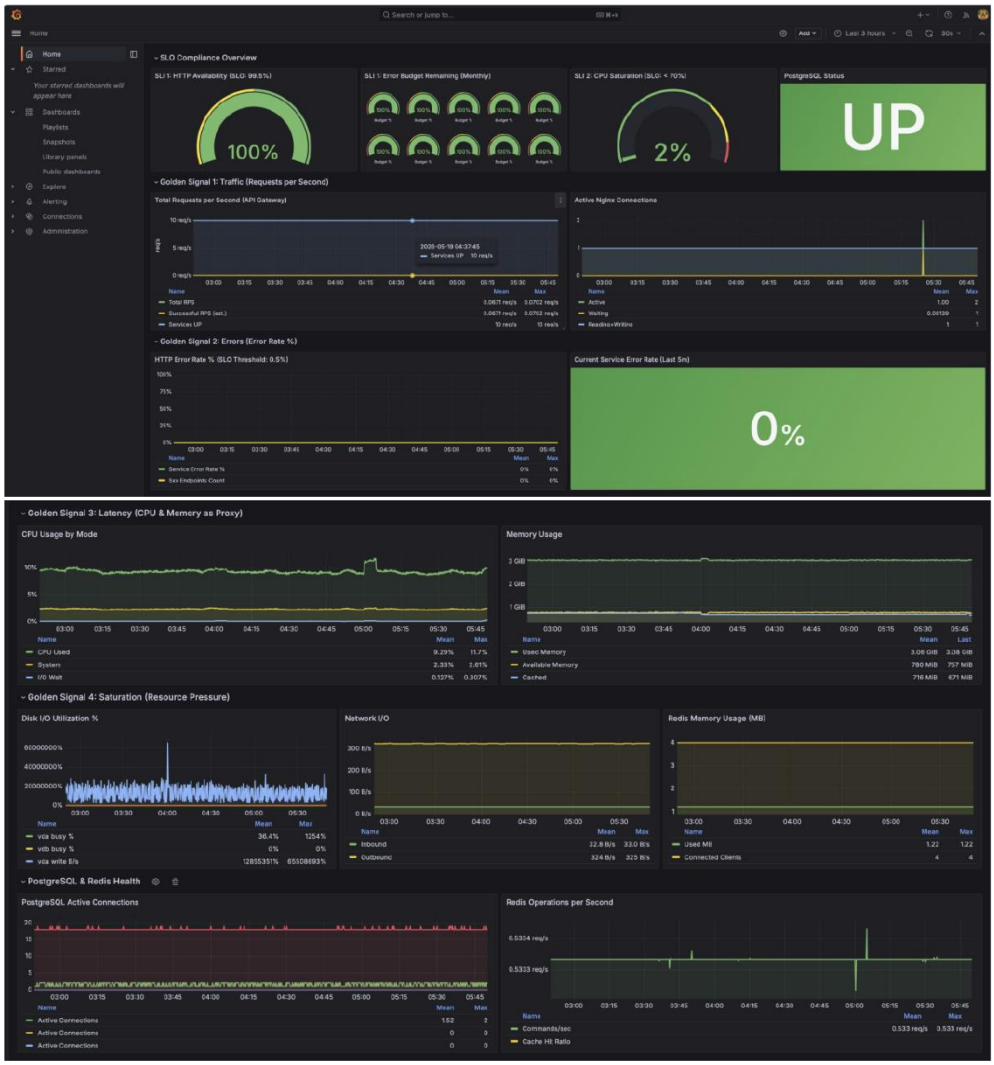
Figure 4: ansible playbooks



Capacity planning
Findings
1. Assessment and payment services show highest burst CPU usage.
2. PostgreSQL is the primary bottleneck under concurrent write load.
3. Gateway remains stable but depends on backend readiness.

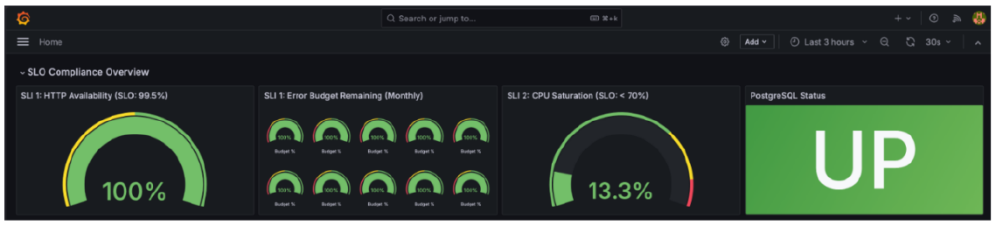
SLI 1: HTTP Availability (SLO: 99.5%) - gauge showing 100%
Query: avg(probe_success{job='edulms-services'}) * 100
Thresholds: red < 99%, yellow >= 99%, green >= 99.5%
- SLI 1: Error Budget Remaining (Monthly) - gauge showing 100%
Query: clamp_min(clamp_max(100-((1-
avg_over_time(probe_success[1h]))/0.005*100),100),0)
Thresholds: red < 25%, yellow >= 25%, green >= 50%
- SLI 2: CPU Saturation (SLO: < 70%) - gauge showing 13.3%
Query: node_load1 / count(node_cpu_seconds_total{mode='idle'})
Thresholds: green < 70%, yellow >= 70%, red >= 90%

Figure 5: cicd workflow yml



Automation

Figure 6: grafana dashboard



Scaling Strategy
1. Horizontal scaling
- Increase replicas for assessment/payment from 2 to 3 during peak windows.
- Keep auth/user/course at 2 replicas baseline.

2. Vertical scaling
- Increase DB node memory class when sustained memory > 75%.

3. Database optimization
- Add indexes on high-write/high-filter columns.
- Review connection pool sizing per service.

Operational Triggers
- CPU > 70% for 10m: scale service replicas +1
- Error rate > 1% for 5m: incident workflow + rollback check
- DB latency p95 > 200ms for 10m: optimize queries / increase DB resources

Next Iteration
- Add automated load test (k6/Locust) before releases.
- Add Kubernetes HPA once full K8s runtime is production-ready.

Figure 7: grafana slo

1. I verified the initial state - all alerts were inactive:
curl http://localhost:9090/api/v1/alerts -> empty array
2. I stopped the postgres-exporter container:
docker stop diplom-postgres-exporter-1
3. Immediately: up{job='postgres-exporter'} dropped to 0
Alert entered PENDING state (waiting for 'for: 1m' duration)
4. After 65 seconds the alert transitioned PENDING -> FIRING
5. I took a screenshot of Prometheus /alerts showing the FIRING state
6. I restored the exporter afterwards:
docker start diplom-postgres-exporter-1
Note: PostgreSQL itself was never stopped - the database continued serving
all application queries throughout the entire demonstration.

Monitoring/alerting

```
PS D:\AITU\sre\final_edulms\educationalLMS> dir observability\prometheus
>> dir observability\grafana\dashboards
    Directory: D:\AITU\sre\final_edulms\educationalLMS\observability\prometheus


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        19.05.2026     03:56           5268 alert_rules.yml
-a----        19.05.2026     03:56           1891 prometheus.yml


    Directory: D:\AITU\sre\final_edulms\educationalLMS\observability\grafana\dashboards


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        19.05.2026     03:56          20834 edulms-golden-signals.json
```

Figure 8: alerts firing

Alert transition: **inactive → pending → FIRING (~65s)**.

alert live

---

9. SLIs & SLOs

Service	Availability	p95 latency	Error rate
Auth	≥ 99.5 %	≤ 150 ms	≤ 0.5 %
Course	≥ 99 %	≤ 200 ms	≤ 1 %
Assessment	≥ 99 %	≤ 250 ms	≤ 1 %
Payment	≥ 99 %	≤ 200 ms	≤ 1 %
User Profile	≥ 99 %	≤ 150 ms	≤ 1 %
Gateway	≥ 99.9 %	≤ 50 ms	≤ 0.1 %

Full SLI/SLO design + error budget policy: `sre/docs/SLI_SLO.md`.

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10. Capacity Planning (summary)

- Assessment + payment services have the highest burst CPU.
- PostgreSQL is the primary bottleneck under concurrent writes.
- HPA in K8s set to scale on CPU 70 %.

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11. Deliverables Checklist

- ☒ 6+ microservices
- ☒ Docker Compose / Swarm
- ☒ Kubernetes manifests
- ☒ Terraform (DigitalOcean)
- ☒ Ansible playbooks
- ☒ **CI/CD GitHub Actions pipeline** (new)
- ☒ Prometheus + Grafana + alerts
- ☒ Incident + postmortem
- ☒ Deployed at https://aitbek.tech
- ☒ Git link: https://github.com/Newterios/educationalLMS

Repository: https://github.com/Newterios/educationalLMS **Live system:** https://aitbek.tech